

“We also noticed differences in the textual content posted by malicious and benign pages. We then built an engine to automatically identify malicious pages from the benign ones. We used bag-of-words based neural network model which could automatically identify malicious pages with over 84 per cent accuracy, looking at only a hundred posts worth of textual content published by a page”

Though the research is based on the last 100 posts of a particular page, it can be extended to the entire history of that page, depending on how much computing power is available. Plus, since malicious pages might switch to sharing benign stuff

levels being even five times greater than the acceptable safe levels set by WHO.

Prof. Pradeep T and his team from Madras have developed a nanotechnology based water purifier that can allow the provision of arsenic free water at affordable rates in any arsenic affected region in the country. His group found that noble metal nanoparticles can completely degrade halocarbons into amorphous carbon and metal halides. This discovery has led to the world's first nanochemistry based filter for pesticide removal. His group developed several technologies to remove contaminants from drinking water. Combining several such materials,



The mooKIT team - Front(L-R): Revathy, Deepak, Tulika. Back(L-R): Sunil, Ravi, Prabhakar, Gaurav

every now and then, a self adaptive model is needed. As a whole, this method sets an exciting precedent for automatic content monitoring and filtering.

ARSENIC PURIFICATION

Availability of potable water has been one of the most persistent problems in India. And with growing industrialisation and lack of proper regulation, even pristine rural landscapes are falling into the categories of polluted areas. One such significant type of pollution is arsenic pollution in water bodies. In fact, Arsenic pollution in the Ganga- Brahmaputra fluvial plains in India and Padma-Meghna fluvial plains in Bangladesh and its consequences to the human health have been reported as one of the world's biggest natural ground-water calamities known to mankind, with

an all-inclusive affordable drinking water purifier has been developed. This technology is being implemented now in the arsenic affected regions of India. About 400,000 people are benefitted from these installations. To take such technologies forward, a company has been incubated with the participation of IIT Madras. This activity is now being expanded globally.

This technology was successfully implemented last year in West Bengal providing arsenic free water for just 5 paise per litre. “We have developed new materials with which almost all health-related contaminants in water can be removed to the required extent. This means that we can deliver quality water effectively. It has been possible to do this at affordable cost as all we do is to pass the contaminated water through a filter.” explains Prof. Pradeep.

He continues, “Our research has already touched several millions of people. Over ten years ago we introduced a technology to remove pesticides from drinking water. That technology is estimated to have reached about 7.5 million people. Our arsenic technology has reached about 0.5 million people. It has been approved for national level implementation. More installations will come up soon.”

NANOFLUIDIC PARTICLES

Medical devices are gradually getting smaller. Today, many of them are portable and also accurate (something we couldn't have in the past). However, the race is on to go even smaller – to the nano level! One such project is by Surjendu Maity, Research Scholar, IIT Guwahati. As he puts it, “There are many advantages of using nanofluidic channels in medical sensing devices. For micro sensing devices, the volume of sample is in the order of a microliter as opposed to the much larger volumes taken for analysis of diseases or biomarkers of diseases. When it comes to the nanochannel, the volume is in the order of a nanoliter or picoliter! We are literally taking molecules instead of wasting blood.” He goes on to add, “Even some rare diseases which are otherwise hard to detect, can be detected using this low cost, accurate and portable nanofluidic point of care testing device. It's so small that it can be integrated into the electronics at chip formation and embedded in a wrist watch, or spectacles for example.” While the exact details of this work and its progress are still under wraps, Surjendu did tell us a prototype should be ready in about 5-10 years.

In order to picture what this means, imagine a micro-syringe which will be integrated with a smartband or a smart-watch perhaps. That nano-tech based syringe will extract your blood without you even noticing, and monitor your health to warn you of any problems. You could set this to also upload reports regularly to your doctor so that he can alert you if certain levels are too high or too low. The only problem is that to manufacture such devices, you need high-end instruments (Electron Beam Lithographs, Mask Writers, Spin Coaters, FESEM, etc.) which are very expensive investments. It might be worth it if the government invested in these in order to help us become a healthier country. 